Project Design and Requirement Analysis Phases:Full Templates

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Project name	Import Data using Transform Maps (Spreadsheet)

PROJECT DESIGN&REQUIREMENT ANALYSIS PHASE: FULL TEMPLATE

- *Project Name:* Import Data using Transform Maps (Spreadsheet)

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1. Technology Stack (Architecture & Stack)

Table-1: Components&Technologies

Component | Technology / Description

User Interface | Web UI / Spreadsheet Management Dashboard (HTML, CSS, JavaScript / React JS)

Application Logic-1 | Backend Processing&Transform Map Engine (Python / Java)

Application Logic-2 | Spreadsheet Parsing&Data Validation Service (Apache POI / Pandas)

Database | Relational Database for Stored Maps and Imported Data (MySQL / PostgreSQL)

Cloud Database / Storage | Cloud Storage for uploaded spreadsheets (AWS S3 / Google Cloud Storage)

Infrastructure | Local Server / Cloud Deployment (Docker, Kubernetes)

Table-2: Application Characteristics

Characteristic | Description

Open-Source Frameworks | Spring Boot / Flask / React

Security Implementations | Role-Based Access Control (RBAC), Input Sanitization, HTTPS / SSL Encryption

Scalable Architecture | 3-Tier Architecture designed for handling bulk data processing

Availability&Performance | Optimized database indexing and caching mechanisms for faster row processing per second

2. Solution Requirements (Functional&Non-Functional)

Functional Requirements (FR):

- *FR-1 (User Authentication):* Registration through Form, Login via Admin credentials.

- *FR-2 (Spreadsheet Upload):* Upload excel/CSV data sheets into the staging table.

- *FR-3 (Transform Map Configuration):* Mapping source spreadsheet columns to target system fields.

- *FR-4 (Data Execution):* Running the transform map to successfully import and clean data.

Non-Functional Requirements (NFR):

- *NFR-1 (Usability):* Simple and intuitive dashboard interface for smooth field mapping.

- *NFR-2 (Security):* Secure file uploads and encrypted user data handling.

- *NFR-3 (Reliability):* Accurate data conversion without data loss or corruption during mapping.

- *NFR-4 (Performance):* Ability to process large datasets quickly with minimal latency.

- *NFR-5&NFR-6 (Availability&Scalability):* Highly available architecture to handle concurrent bulk file imports.

3. Data Flow Diagrams&User Stories

Data Flow Overview:

1. User uploads a spreadsheet file containing raw data into the system.

2. The application parses the spreadsheet structure and loads it into a staging area.

3. The Transform Map configures and transforms source fields into target fields.

4. Processed data is securely stored in the main database and confirmation is displayed to the user.

User Stories Template:

User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance Criteria | Priority | Release

Administrator | Spreadsheet Upload | USN-1 | As an admin, I can upload spreadsheet files (CSV/Excel) to the system. | File is validated and loaded into staging. | High | Sprint-1

Administrator | Transform Mapping | USN-2 | As an admin, I can create and configure Transform Maps between source and target fields. | Mapping rules are saved successfully. | High | Sprint-1

Administrator | Data Execution | USN-3 | As an admin, I can run the transform map to import data into target tables. | Data is imported accurately without errors. | High | Sprint-2

User | Report Generation | USN-4 | As a user, I can view import logs and error reports after execution. | Error summary logs are displayed clearly. | Medium | Sprint-2